The Isometrically Homogeneous Hilbert-Space Problem

Audit of the Supplied Argument, Literature Status, and Structural Reductions

Research report prepared from the supplied source paper and partial solution

22 June 2026

Status. The supplied conditional theorem is correct. Its stated "symmetric distortion" corollary needs a small square-root normalization in its proof, but its qualitative conclusion is correct. I found no published full proof or counterexample in the literature search described below. The problem therefore remains open on the evidence available to this report. Sections 5–9 give additional rigorous reductions: an optimal Hilbert position exists; every infinite-dimensional subspace attains the full optimal distortion range; optimal Hilbert structures are unique modulo compact operators; every linear isometric embedding is a compact perturbation of a Hilbert isometry; and a hypothetical counterexample admits a pure unilateral-shift normal form.

Abstract

Let X be a separable infinite-dimensional Banach space linearly isometric to every one of its closed infinite-dimensional subspaces. Cúth, Doležal, Doucha and Kurka asked whether X must be isometric to ℓ_2. The isomorphic homogeneous-space theorem implies only that X is isomorphic to ℓ_2, and arbitrary distortion of Hilbert space prevents the usual shortcut from isomorphism to almost-isometry.

This report first verifies the supplied almost-Hilbertian stabilization argument. It then records the literature status and develops a sequence of intrinsic reductions. If $D = d_{\mathrm{BM}}(X, \ell_2)$, the infimum defining D is attained by an equivalent Hilbert norm $|\cdot|_q$ normalized by $|x|_q \leq \|x\| \leq D|x|_q$. In a non-Hilbert counterexample, every closed infinite-dimensional subspace has norm-ratio infimum 1 and supremum D in this same Hilbert position. Any other normalized optimal Hilbert norm differs from q by a compact positive perturbation. Consequently, every linear $\|\cdot\|$-isometric embedding $U : X \to X$ satisfies $U^*U - I$ compact with respect to q, and hence is a compact perturbation of a Hilbert isometric embedding. Finally, after passing to an isometric subspace, one may arrange that X is an equivalent renorming of ℓ_2 for which the canonical right shift is simultaneously a Banach-space isometry and a Hilbert isometry. These conclusions sharply constrain a counterexample but do not yet force $D = 1$.

Contents

Packet status and source evidence

This is a partial-result packet for arXiv:2204.06834, Question 3 of Cúth–Doležal–Doucha–Kurka. It does not claim a full solution or a counterexample. The source problem remains open on the evidence reviewed here. The contribution of this upgraded packet is to preserve the verified almost-Hilbertian obstruction from the previous packet, correct the square-root normalization in its symmetric-distortion corollary, and record additional reductions that any non-Hilbert counterexample must satisfy.

- Previous active packet preserved at `history/2026-06-22_previous_almost_hilbertian_partial/`.
- Supplied upgrade report preserved in the dated evidence folder.
- Source paper preserved as `source_paper.pdf`.

Question 2. Is $\langle \mathbb{G} \rangle_{\simeq}$ a G_δ set in $\mathcal{P}_\infty$ or in $\mathcal{B}$? Is it at least Borel?

Solving the homogeneous Banach space problem, Komorowski and Tomczak-Jaegermann ([35]), and Gowers ([29]) proved that if a separable infinite-dimensional Banach space is isomorphic to all of its closed infinite-dimensional subspaces, then it is isomorphic to ℓ_2. It seems that the isometric variant of this result is open; that is, whether ℓ_2 is the only (up to isometry) separable infinite-dimensional Banach space that is isometric to all of its infinite-dimensional closed subspaces. We note that any Banach space satisfying this criterion must be, by the Gowers' result, isomorphic to ℓ_2. Our initial interest in this problem was that we observed that a positive answer implies that whenever $\langle X \rangle_{\equiv}$ is closed in $\mathcal{P}_\infty$ then $X \equiv \ell_2$. Eventually we found another argument (see Subsection 2.1), but the question is clearly of independent interest.

Question 3. Let X be a separable infinite-dimensional Banach space which is isometric to all of its closed infinite-dimensional subspaces. Is then X isometric to ℓ_2?

We note here that Question 3 was already attacked by N. de Rancourt [15], who was able to prove that if X is as above (that is, isometric to all of its closed infinite-dimensional subspaces) then, for every $\varepsilon > 0$, X admits a $(1+\varepsilon)$-unconditional basis.

1 The problem and terminology

Throughout, $\mathbb{F}$ denotes either $\mathbb{R}$ or $\mathbb{C}$. A Banach space $X = (X, \|\cdot\|)$ is called *isometrically homogeneous* in this report if, for every closed infinite-dimensional subspace $Y \subseteq X$, there is a surjective linear isometry $U : X \to Y$.

The question considered by Cúth, Doležal, Doucha and Kurka is the following.

Question 1.1 (Isometric homogeneous-space problem)**.** If a separable infinite-dimensional Banach space is isometrically homogeneous, must it be linearly isometric to ℓ_2?

The homogeneous-space theorem of Komorowski–Tomczak-Jaegermann and Gowers implies that such an X is linearly *isomorphic* to ℓ_2. Thus one may fix an equivalent Hilbert norm on the same vector space. The open issue is whether the given norm itself satisfies the parallelogram identity.

For two isomorphic Banach spaces E, F, write

$$d_{\mathrm{BM}}(E, F) = \inf\{\|T\| \, \|T^{-1}\| : T : E \to F \text{ is a linear isomorphism}\}.$$

When F is a separable infinite-dimensional Hilbert space, we abbreviate $d_{\mathrm{BM}}(E, F)$ to $d(E, \ell_2)$.

2 Literature status

The journal version of the source paper appeared in 2024 and still lists Question 1.1 as open [2]. The strongest directly targeted result I found is due to de Rancourt:

Theorem 2.1 (de Rancourt [3])**.** *If X is isometrically homogeneous, then for every $\varepsilon > 0$ the space X admits a $(1 + \varepsilon)$-unconditional basis.*

De Rancourt explicitly identifies the missing ingredient as a sufficiently strong quantitative form of the Komorowski–Tomczak-Jaegermann theorem. The obstruction is genuine. Odell and Schlumprecht proved that ℓ_2 is arbitrarily distortable: for every $\lambda > 1$ there is an equivalent norm on ℓ_2 whose restriction to every infinite-dimensional subspace has oscillation exceeding λ relative to the original Hilbert norm [6]. They also constructed, for prescribed $\lambda > 1$, equivalent norms with no λ-unconditional basic sequence. Thus Theorem 2.1 rules out the most hostile known distortions, but it does not by itself supply an almost-Euclidean infinite-dimensional restriction.

A second warning comes from Casazza's survey [1]: exact recurrence of all finite-dimensional isometry types in every infinite-dimensional subspace is not sufficient in arbitrary Banach spaces; c_0 has that recurrence property. Any proof of Question 1.1 must therefore use the additional Hilbertian isomorphism structure, not only finite-dimensional recurrence.

The search performed for this report covered the exact wording of the question, variants of "isometrically homogeneous Banach space", the title and citations of the source paper, citations of de Rancourt's paper, and arXiv/journal material through 22 June 2026. No later full proof or counterexample was located. This is evidence about the indexed literature, not a claim that an unpublished or obscure result cannot exist.

3 Audit of the supplied partial result

The central theorem in the supplied packet is correct. It is useful to state it in a slightly normalized form.

Theorem 3.1 (Verified supplied theorem)**.** *Let $X = (X, \|\cdot\|)$ be separable, infinite-dimensional and isometrically homogeneous. Suppose that there is an equivalent Hilbert norm $|\cdot|$ such that for every $\varepsilon > 0$ there are a closed infinite-dimensional subspace $Y \subseteq X$ and $a > 0$ satisfying*

$$a|y| \le \|y\| \le (1+\varepsilon)a|y| \qquad (y \in Y).$$

Then X is linearly isometric to ℓ_2.

Proof. Fix a finite-dimensional subspace $F \subseteq X$ and $\varepsilon > 0$. Choose Y and a as in the hypothesis, and let $U : X \to Y$ be a surjective linear $\|\cdot\|$-isometry. For $x \in F$,

$$a|Ux| \le \|Ux\| = \|x\| \le (1+\varepsilon)a|Ux|.$$

Since $(U(F), |\cdot|)$ is Euclidean, $d_{\mathrm{BM}}(F, \ell_2^{\dim F}) \le 1+\varepsilon$. Letting $\varepsilon \downarrow 0$ gives distance one. In finite dimension, distance one is attained: after normalizing almost-isometries, compactness of the unit sphere in the relevant operator space yields an actual linear isometry. Therefore every finite-dimensional subspace of X is Euclidean. In particular every two-dimensional subspace satisfies the parallelogram law, so the Jordan–von Neumann theorem makes $\|\cdot\|$ an inner-product norm. Separable infinite-dimensional Hilbert spaces are linearly isometric to ℓ_2. □

There is one normalization point in the packet's corollary. For a fixed Hilbert norm $|\cdot|$ and an infinite-dimensional subspace Y, put

$$m_Y = \inf_{|y|=1,\, y\in Y} \|y\|, \qquad M_Y = \sup_{|y|=1,\, y\in Y} \|y\|.$$

Then

$$\inf_{a>0} \sup_{0\ne y\in Y} \max\left\{\frac{\|y\|}{a|y|}, \frac{a|y|}{\|y\|}\right\} = \sqrt{\frac{M_Y}{m_Y}}. \tag{1}$$

Indeed the left side is $\inf_{a>0} \max\{M_Y/a, a/m_Y\}$, minimized at $a = \sqrt{M_Y m_Y}$. The theorem's hypothesis is the ratio condition $M_Y/m_Y \le 1+\varepsilon$, whereas the symmetric expression in (1) is its square root.

Corollary 3.2 (Corrected proof of the packet's corollary)**.** *If a non-Hilbert counterexample to Question 1.1 exists, then for every equivalent Hilbert norm $|\cdot|$ there is $\delta > 0$ such that every closed infinite-dimensional $Y \subseteq X$ satisfies*

$$\inf_{a>0} \sup_{0\ne y\in Y} \max\left\{\frac{\|y\|}{a|y|}, \frac{a|y|}{\|y\|}\right\} \ge 1+\delta.$$

Proof. Negating the stabilization hypothesis in Theorem 3.1 gives an $\eta > 0$ such that $M_Y/m_Y > 1+\eta$ for every infinite-dimensional Y. Equation (1) then gives a uniform lower bound strictly larger than $\sqrt{1+\eta}$; decreasing the bound slightly yields the stated non-strict inequality. Thus the qualitative conclusion in the packet is correct, but the square-root conversion should be made explicit. □

4 An intrinsic almost-Hilbert criterion

The fixed ambient Hilbert norm in Theorem 3.1 is unnecessary. The real mechanism is intrinsic Banach–Mazur stabilization.

Theorem 4.1 (Intrinsic criterion)**.** *Let X be isometrically homogeneous. If*

$$\inf\{d(Y,\ell_2) : Y \subseteq X \text{ closed and infinite-dimensional}\} = 1,$$

then X is linearly isometric to ℓ_2.

Proof. Fix a finite-dimensional $F \subseteq X$ and $\varepsilon > 0$. Choose an infinite-dimensional $Y \subseteq X$ with $d(Y,\ell_2) < 1+\varepsilon$, and let $U : X \to Y$ be a surjective $\|\cdot\|$-isometry. Restricting an $(1+\varepsilon)$-Hilbertian structure on Y to $U(F)$ shows $d(F,\ell_2^{\dim F}) < 1+\varepsilon$. The rest is exactly the compactness and parallelogram argument from Theorem 3.1. □

Remark 4.2. Theorem 4.1 isolates the precise gap left by arbitrary distortion. A full positive solution would follow from proving that an isometrically homogeneous renorming of ℓ_2 cannot remain a fixed positive distance from Hilbert space on every infinite-dimensional subspace.

5 Optimal Hilbert position

From now on let $X = (H, \|\cdot\|)$ be isomorphic to a separable infinite-dimensional Hilbert space, and set

$$D = d(X,\ell_2).$$

The first useful point is that, although Banach–Mazur infima need not be attained for arbitrary pairs of infinite-dimensional spaces, the Hilbert target permits attainment.

Theorem 5.1 (Attainment of the Hilbert distance)**.** *There is an equivalent Hilbert norm $|\cdot|_q$ on H such that*

$$|x|_q \le \|x\| \le D|x|_q \qquad (x \in H). \tag{2}$$

In particular, if $D = 1$, then $\|\cdot\|$ is a Hilbert norm.

Proof. Fix a reference Hilbert norm $|\cdot|_0$. For each k choose a Hilbert norm $|\cdot|_k$, scaled so that

$$|x|_k \le \|x\| \le (D+k^{-1})|x|_k.$$

Write $|x|_k^2 = \langle A_k x, x\rangle_0$ with A_k positive and invertible. Equivalence of $|\cdot|_0$ and $\|\cdot\|$, together with the displayed inequalities, gives constants $0 < c < C < \infty$ independent of k such that $cI \le A_k \le CI$. The order interval $[cI, CI]$ is compact in the weak operator topology after passing to a subnet (or, on separable H, a subsequence). Let $A_k \to A$ weakly. Then $cI \le A \le CI$ and $|x|_q^2 := \langle Ax, x\rangle_0$ defines an equivalent Hilbert norm. Passing to the limit pointwise in the two inequalities gives (2). □

Now impose isometric homogeneity.

Theorem 5.2 (Hereditary full distortion)**.** *Assume X is isometrically homogeneous and let q be normalized as in* (2)*. For every closed infinite-dimensional $Y \subseteq H$,*

$$\inf_{\substack{y\in Y\\ |y|_q=1}} \|y\| = 1, \qquad \sup_{\substack{y\in Y\\ |y|_q=1}} \|y\| = D. \tag{3}$$

Proof. Let the two quantities be m_Y and M_Y. From (2), $1 \le m_Y \le M_Y \le D$. Since Y is isometric to X, $d(Y,\ell_2) = D$. The identity between $(Y, |\cdot|_q)$ and $(Y, \|\cdot\|)$ has distortion M_Y/m_Y, so

$$D \le M_Y/m_Y \le D.$$

Hence $M_Y = Dm_Y$. The bounds $M_Y \le D$ and $m_Y \ge 1$ force $m_Y = 1$ and $M_Y = D$. □

Corollary 5.3. *A non-Hilbert counterexample has $D > 1$ and, in an optimal Hilbert position, its entire distortion interval $[1, D]$ is unavoidable on every infinite-dimensional subspace. In particular, neither endpoint can be removed by passing to a subspace.*

This is stronger than the supplied corollary because it uses an optimal Hilbert norm and identifies the exact hereditary endpoint values.

6 Rigidity of Hilbert structures modulo compact operators

Let q be an optimal Hilbert norm as above. If r is another Hilbert norm, write

$$|x|_r^2 = \langle Ax, x\rangle_q$$

for its positive invertible comparison operator A. The next result converts hereditary full distortion into an essential-spectrum constraint.

Theorem 6.1 (Near-optimal essential-spectrum rigidity)**.** *Assume X is isometrically homogeneous and q satisfies* (2)*. Let r be a Hilbert norm normalized by*

$$|x|_r \le \|x\| \le D'|x|_r \qquad (x \in H),$$

where $D' \ge D$. If $|x|_r^2 = \langle Ax, x\rangle_q$, then

$$\sigma_{\mathrm{ess}}(A) \subseteq \left[\left(\frac{D}{D'}\right)^2, 1\right]. \tag{4}$$

Proof. Suppose first that the essential spectrum meets $(1, \infty)$. Then for some $\eta > 0$ the spectral subspace $Y = \mathbf{1}_{[1+\eta,\infty)}(A)H$ is infinite-dimensional. On Y, $|y|_r \ge \sqrt{1+\eta}|y|_q$, and $|y|_r \le \|y\|$. Therefore $\|y\|/|y|_q \ge \sqrt{1+\eta}$ on $Y \setminus \{0\}$, contradicting the first identity in (3).

Now suppose the essential spectrum meets $(0, (D/D')^2)$. Choose $c < D/D'$ so that $Y = \mathbf{1}_{(0,c^2]}(A)H$ is infinite-dimensional. On Y, $|y|_r \le c|y|_q$, whence

$$\|y\| \le D'|y|_r \le D'c|y|_q < D|y|_q,$$

contradicting the second identity in (3). This proves (4). $\square$

Corollary 6.2 (Compact uniqueness of optimal Hilbert positions)**.** *If q_1 and q_2 are both normalized optimal Hilbert norms,*

$$|x|_{q_i} \le \|x\| \le D|x|_{q_i} \quad (i = 1, 2),$$

and $|x|_{q_1}^2 = \langle Ax, x\rangle_{q_2}$, then $A - I$ is compact.

Proof. Apply Theorem 6.1 with $D' = D$. It gives $\sigma_{\mathrm{ess}}(A) \subseteq \{1\}$. The essential spectrum of a bounded self-adjoint operator on an infinite-dimensional Hilbert space is nonempty, so it equals $\{1\}$. Equivalently, $A - I$ is compact. $\square$

7 Isometries are Hilbertian modulo compacts

The preceding compact uniqueness applies directly to every into-isometry. Adjoints and compactness in this section are taken with respect to the optimal Hilbert norm q.

Theorem 7.1 (Compact polar rigidity)**.** *Let $U : X \to X$ be a linear $\|\cdot\|$-isometric embedding, not necessarily surjective. Then*

$$U^*U - I \quad \text{is compact.} \tag{5}$$

Consequently, in the polar decomposition $U = V|U|$, the operator V is a q-Hilbert isometric embedding with the same range as U, and

$$U - V \quad \text{is compact.}$$

If U is surjective, V is unitary.

Proof. Define $|x|_{q_U} = |Ux|_q$. Since U is a $\|\cdot\|$-isometry,

$$|x|_{q_U} = |Ux|_q \le \|Ux\| = \|x\| \le D|Ux|_q = D|x|_{q_U}.$$

Thus q_U is another normalized optimal Hilbert norm. Its comparison operator with q is U^*U, so Corollary 6.2 gives (5). Functional calculus yields $|U| = (U^*U)^{1/2} = I + K$ with K compact. Hence $U = V(I + K)$ and $U - V = VK$ is compact. Since $|U|$ is invertible, U and V have the same range. □

Proposition 7.2 (Dual quotient homogeneity)**.** *If X is isometrically homogeneous, then every infinite-dimensional quotient of X^* is linearly isometric to X^*.*

Proof. The homogeneous-space theorem gives $X \simeq \ell_2$, so X is reflexive. Let $Z \subseteq X^*$ be closed and suppose X^*/Z is infinite-dimensional. Because X is reflexive, norm-closed subspaces of X^* are weak-star closed. With $Y = Z_\perp \subseteq X$, one has $Z = Y^\perp$, and restriction gives the canonical quotient isometry

$$X^*/Z = X^*/Y^\perp \cong Y^*.$$

The quotient is infinite-dimensional exactly when Y is. Homogeneity gives $Y \cong X$ isometrically, hence $Y^* \cong X^*$ isometrically. □

8 Interaction with near-unconditional bases

Theorem 2.1 can be combined with Theorem 6.1 to obtain a precise "unitary modulo compacts" form of the sign symmetries. This does not solve the problem because the basis may vary with ε, but it narrows the remaining quantitative gap.

Proposition 8.1 (Essential Hilbertianity of sign changes)**.** *Let (x_n) be a K-unconditional basis of X, using the sign-change convention: every coordinate sign operator S_θ satisfies $\|S_\theta\|_{X\to X} \le K$. Relative to an optimal Hilbert norm q,*

$$\sigma_{\mathrm{ess}}(S_\theta^* S_\theta) \subseteq [K^{-2}, K^2]. \tag{6}$$

Moreover, there is a q-unitary W_θ such that

$$\|S_\theta - W_\theta\|_{\mathrm{ess},q} \le K - 1. \tag{7}$$

Proof. Since $S_\theta^{-1} = S_\theta$, unconditionality gives $K^{-1}\|x\| \le \|S_\theta x\| \le K\|x\|$. Define

$$|x|_r = K^{-1}|S_\theta x|_q.$$

Then $|x|_r \le \|x\|$ and

$$\|x\| \le K\|S_\theta x\| \le KD|S_\theta x|_q = K^2 D|x|_r.$$

The comparison operator of r with q is $A = K^{-2}S_\theta^* S_\theta$. Applying Theorem 6.1 with $D' = K^2 D$ gives $\sigma_{\mathrm{ess}}(A) \subseteq [K^{-4}, 1]$, which is (6).

Let $S_\theta = W_\theta|S_\theta|$ be its Hilbert polar decomposition. Because S_θ is invertible, W_θ is unitary. Equation (6) implies $\sigma_{\mathrm{ess}}(|S_\theta|) \subseteq [K^{-1}, K]$, so

$$\||S_\theta| - I\|_{\mathrm{ess},q} \le \max\{K-1, 1-K^{-1}\} = K-1.$$

Finally $S_\theta - W_\theta = W_\theta(|S_\theta| - I)$. □

Corollary 8.2. *For an isometrically homogeneous space and every $\varepsilon > 0$, there is a basis whose entire sign-change group is, in the Calkin algebra of the optimal Hilbert space, within ε of unitary operators.*

Remark 8.3. The unresolved step is to turn these basis-dependent asymptotic unitary symmetries into a single infinite-dimensional subspace on which $\|\cdot\|/|\cdot|_q$ has oscillation tending to one. Arbitrary distortion shows that this conclusion cannot be obtained from the Hilbertian isomorphism alone, while Proposition 8.1 shows exactly what additional operator-theoretic information de Rancourt's theorem provides.

9 A pure unilateral-shift normal form

A counterexample, if it exists, may be put in a particularly rigid self-similar position. The construction uses a shift-invariant Banach limit and the Wold decomposition.

Theorem 9.1 (Shift normal form)**.** *Let X be isometrically homogeneous. There is a Banach space isometric to X, represented as $(\ell_2, \|\cdot\|)$, and an optimal Hilbert norm equal to the standard ℓ_2 norm, such that the canonical right shift*

$$S(a_1, a_2, \ldots) = (0, a_1, a_2, \ldots)$$

is simultaneously a $\|\cdot\|$-isometry and a Hilbert isometry. The represented space remains isometrically homogeneous.

Proof. Choose a closed hyperplane $H_1 \subset X$ and a surjective $\|\cdot\|$-isometry $T : X \to H_1$, viewed as an into-isometry $T : X \to X$. Start with an optimal Hilbert norm $|\cdot|_0$ satisfying $|x|_0 \le \|x\| \le D|x|_0$. Since T is a Banach-space isometry, for every $n \ge 0$,

$$D^{-1}|x|_0 \le |T^n x|_0 \le D|x|_0.$$

Let LIM be a shift-invariant Banach limit and define

$$|x|_T^2 = \mathrm{LIM}_{n\to\infty} |T^n x|_0^2.$$

This is a positive definite quadratic form equivalent to $|\cdot|_0$; the parallelogram identity passes through the Banach limit. Shift invariance gives $|Tx|_T = |x|_T$. More precisely, because $|T^n x|_0 \le \|T^n x\| = \|x\|$ and $\|x\| = \|T^n x\| \le D|T^n x|_0$, one has

$$|x|_T \le \|x\| \le D|x|_T.$$

Thus $|\cdot|_T$ is still optimal.

Apply the Wold decomposition to the Hilbert isometry T:

$$H = H_u \oplus \bigoplus_{n\ge 0} T^n W, \qquad W = H \ominus T(H).$$

Since $T(H)$ has codimension one, $\dim W = 1$. The shift part $H_s = \bigoplus_{n\ge 0} T^n W$ is infinite-dimensional and closed for both $|\cdot|_T$ and $\|\cdot\|$. By homogeneity, $(H_s, \|\cdot\|)$ is isometric to X. Replacing X by this isometric copy removes the unitary part. If w is a Hilbert unit vector spanning W, the sequence $(T^n w)_{n\ge 0}$ is an orthonormal basis of H_s, and under the resulting identification with ℓ_2, T becomes the canonical right shift. Any infinite-dimensional closed subspace of H_s is also one of X, so homogeneity is preserved. □

10 The resulting counterexample profile

Combining the preceding results gives a compact description of what remains to be ruled out.

Theorem 10.1 (Necessary profile of a counterexample)**.** *If Question 1.1 has a negative answer, then there is an equivalent norm $\|\cdot\|$ on ℓ_2 and a number $D > 1$ with all of the following properties.*

(i) $|x|_2 \le \|x\| \le D|x|_2$, *and* $D = d((\ell_2, \|\cdot\|), \ell_2)$.

(ii) *On every infinite-dimensional closed $Y \subseteq \ell_2$,*

$$\inf_{|y|_2=1,\, y\in Y} \|y\| = 1, \qquad \sup_{|y|_2=1,\, y\in Y} \|y\| = D.$$

(iii) *The canonical right shift is a $\|\cdot\|$-isometry.*

(iv) *Every linear $\|\cdot\|$-isometric embedding U is a compact perturbation of a Hilbert isometric embedding; equivalently, $U^*U - I$ is compact.*

(v) *For every $\varepsilon > 0$, the space has a $(1+\varepsilon)$-unconditional basis, and the associated sign changes satisfy Proposition 8.1.*

(vi) *Every finite-dimensional subspace of X embeds linearly isometrically into every infinite-dimensional subspace of X.*

(vii) *Every infinite-dimensional quotient of X^* is linearly isometric to X^*.*

Proof. Items (i)–(iv) are Theorems 5.1, 5.2, 7.1 and 9.1. Item (v) is de Rancourt's theorem together with Proposition 8.1. For (vi), if $F \subset X$ is finite-dimensional and $Y \subset X$ is infinite-dimensional, a surjective isometry $U : X \to Y$ maps F isometrically into Y. Item (vii) is Proposition 7.2. □

11 Why the remaining step is difficult

The reductions above leave a sharply formulated target: prove that the profile in Theorem 10.1 is impossible when $D > 1$. Several tempting shortcuts fail.

B1. Isomorphism does not give an almost-Hilbert subspace. Odell–Schlumprecht distortion directly blocks this inference.

B2. A 1-unconditional basis does not itself make the norm Euclidean. For example,

$$\|x\|_\alpha = \|x\|_2 + \alpha\|x\|_\infty \qquad (\alpha > 0)$$

is an equivalent non-Hilbert norm on ℓ_2 with a 1-unconditional canonical basis. It is not a counterexample: suitable flat block subspaces are arbitrarily close to Hilbert. The missing issue is to derive such a flat subspace uniformly from the hereditary self-isometry property.

B3. Finite-dimensional recurrence is insufficient by itself. Casazza's c_0 observation shows that every finite-dimensional isometry type can recur in every infinite-dimensional subspace without the ambient norm being Hilbertian.

B4. Compact perturbation is not yet exact rigidity. The Calkin algebra contains many proper isometries and many equivalent infinite projections. Thus Theorem 7.1 is a substantial operator reduction, but Fredholm or Calkin-algebra information alone does not force $D = 1$.

B5. The shift symmetry is one-sided. Shift invariance gives exact self-similarity under insertion of a zero coordinate, but not invariance under arbitrary permutations or Hilbert rotations. A proof must manufacture enough additional symmetry from homogeneity, or show that hereditary full distortion is incompatible with the available near-unitary sign symmetries.

A concrete next lemma that would finish the problem is the following.

Question 11.1 (Quantitative stabilization target)**.** Does every equivalent norm on ℓ_2 satisfying items (ii), (iii), and (v) of Theorem 10.1 have, for every $\varepsilon > 0$, an infinite-dimensional subspace Y with $d(Y, \ell_2) < 1 + \varepsilon$?

A positive answer to Question 11.1, even without using all the other items in the profile, would combine with Theorem 4.1 to solve the original problem.

12 Conclusion

The supplied theorem is sound and identifies a valid sufficient condition. The only correction needed is to distinguish the ratio M_Y/m_Y from the symmetric distortion $\sqrt{M_Y/m_Y}$ in the corollary.

No full answer or counterexample was found in the literature search, and the 2024 journal source still presents the question as open. The additional results in this report do not settle it, but they reduce a hypothetical counterexample to a highly constrained renorming of ℓ_2: it must be hereditarily fully distortive in an attained optimal Hilbert position, possess near-perfect unconditional bases, have all isometric embeddings Hilbertian modulo compact operators, and admit an exact pure unilateral-shift symmetry. The remaining bottleneck is to turn these asymptotic and operator-theoretic symmetries into an almost-Hilbertian infinite-dimensional restriction.

Novelty caution. The deductions in Sections 5–9 were developed and checked for this report. I did not find these exact formulations in the searched literature, but they have not undergone peer review.

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